

Логарифмически вогнутые меры

Homework

1 сентября 2026 года

1. (Newton 1707) Let $p(x) = \sum_{k=0}^n a_k x^k$ have only real roots and nonnegative coefficients. Prove that the sequence $a_k / \binom{n}{k}$ satisfies $c_k^2 \geq c_{k-1} c_{k+1}$, and deduce that (a_k) is log-concave. Check this on $(1+x)^n$ and explain why the binomial coefficients are log-concave.
2. (Ball 1988) For log-concave f with $f(0) > 0$ and $p > 0$ put $K_p(f) = \{x : p \int_0^\infty r^{p-1} f(rx) dr \geq f(0)\}$. Prove that $K_p(f)$ is a convex body and that $K_p(\mathbf{1}_K) = K$ for $0 \in K$.
3. Prove that every log-concave probability measure can be put in isotropic position by an affine map, uniquely up to orthogonal transformations.
4. (Brascamp–Lieb 1976) Let $d\mu = e^{-V} dx$ with $\nabla^2 V \geq \varepsilon I$. Prove $\text{Var}_\mu(g) \leq \varepsilon^{-1} \int |\nabla g|^2 d\mu$ in the Gaussian case $V = \|x\|^2/2$, and look up the general proof.
5. (Borell) Prove Khinchine-type inequalities for log-concave measures: for a linear functional ℓ , $(\int |\ell|^p d\mu)^{1/p} \leq Cp \int |\ell| d\mu$ with an absolute constant. (*Hint: Borell's lemma from the class sheet.*)
6. Prove that log-concavity of a random vector is preserved by affine images, by summing independent vectors and by conditioning on a convex set.
7. (Капачёв 2.2–2.3) (a) n points are chosen uniformly and independently on a circle; find the probability that their convex hull contains the centre. (b) For $n \geq 7$ points chosen uniformly in a disc, which is more likely: that the convex hull is a triangle, or that it is a quadrilateral?

8. (IMC 2025 shortlist) Four points are chosen independently and uniformly in the unit cube $[0, 1]^3$. What is the probability that their convex hull contains the centre of the cube?
9. Let μ be a log-concave measure and A a convex set. Prove that $t \mapsto \mu(tA)$ is log-concave on $(0, \infty)$; in particular $\mu(A)^2 \geq \mu(\frac{1}{2}A)\mu(2A)$. Check this for the Gaussian measure and $A = [-1, 1]$.
10. Deduce Grünbaum's inequality in $\mathbb{R}^n$ from the one-dimensional case plus log-concavity of marginals, and compare with the proof through Brunn's theorem.
11. (Sylvester 1864, the four-point problem) Four points are chosen uniformly and independently in a triangle. Prove that the probability that they are in convex position equals $2/3$, and set up the analogous computation for a disc.
12. Compute the isotropic constant of the cube and of the ball, and check that both are bounded by absolute constants; explain what the slicing problem asks about this quantity.